



Iowa City Robotics

Marshmallow Towers

Summary

Students work together to build the tallest tower they can using spaghetti and marshmallows. While building, they explore how shapes and teamwork help make structures strong.

Description

- Work with your group to build the tallest tower you can using spaghetti and marshmallows
- The tower must stand on its own without anyone holding it

Materials

Material	Quantity
Spaghetti	15 per group of 4
Mini Marshmallows	10 per group of 4
Yardstick	1 per classroom
Copy paper	1 sheet per group

Directions

- Divide students into small groups (3–4 students)
- Explain the challenge
 - You will have 15 minutes to build your tower
 - The tower must stand by itself
 - You can only use the materials given to your group
 - Try to build the tallest tower you can
- Give groups 2–3 minutes to talk about their plan
 - What shape will your tower be?
 - How will you make it strong?
- Give each group 15 spaghetti noodles and 10 marshmallows

- Start the timer and let students build
- When time is up, measure the towers with a yardstick and celebrate the tallest tower

Activity Extension

- Build Again
 - Let groups try a second round using what they learned the first time
- Shape Exploration
 - Ask students to look at their towers and find shapes
 - Do they see triangles, squares, or rectangles?
 - Practice drawing those shapes on paper
- Compression and Tension Activity
 - Have students work with a partner
 - Stand facing each other and gently press palms together like a high five
 - Slowly lean in toward each other. This is called compression (pushing together)
 - Now hold hands and gently lean away from each other. This is called tension (pulling apart)
 - Ask students
 - What do you feel in your arms?
 - Is it pushing or pulling?

Discussion Questions

- What helped your tower stand up?
- What made your tower strong or weak?
- What shapes did you use?
- If you built again, what would you do differently?
- Why do you think some towers fell over?

What is Happening?

- Gravity pulls everything down, including the tower
- The spaghetti noodles help hold the tower up
- Some parts of the tower hold more weight than others
- Triangles are one of the strongest shapes for building because they help spread out the weight